Animex

Multimodal Interactive Technologies and Applications Project

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1 Project Resources and Installation Guide

1.1 Project Repositories

Here you can find the links to the Animex Git repositories.

- Front-end application
- Back-end server

1.2 Usage Guide

To test the Animex application you must follow these steps:

1.2.1 Android

1. Go to the front-end repository.
2. Download Animex.apk from the releases section.
3. Install the APK on your Android device.
4. Run the application.

Fallback: Clone the repository, follow the repo's README to run the program through Android Studio.

1.2.2 iOS

1. Go to the front-end repository.
2. Clone the repository.
3. Follow the steps in the repo's README to run the program through XCode.

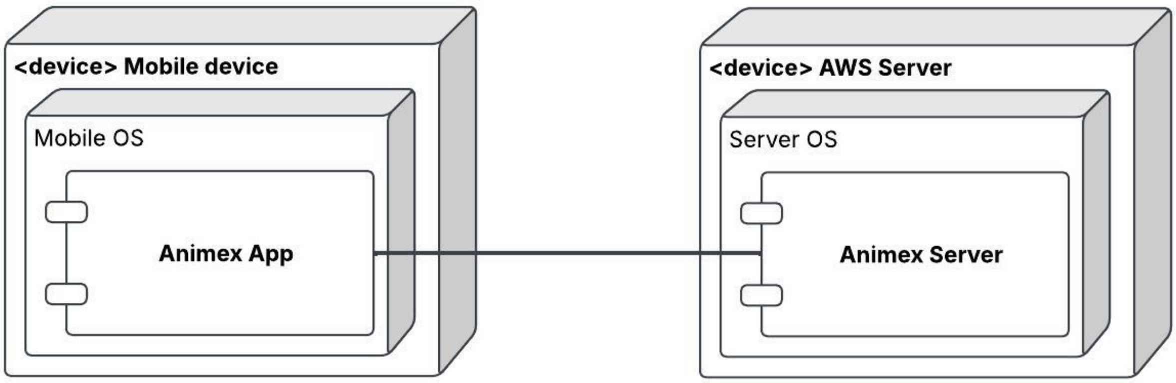
2 Conceptual Description

3 Main Tasks

4 Project Architecture

The Animex project is split across a two-tier architecture:

- **Front-end Mobile Application**: handles user input and displays graphical interface.
- **Back-end Server**: receives and sanitizes requests, prompts *Google Cloud*'s APIs and elaborates results for the front-end.



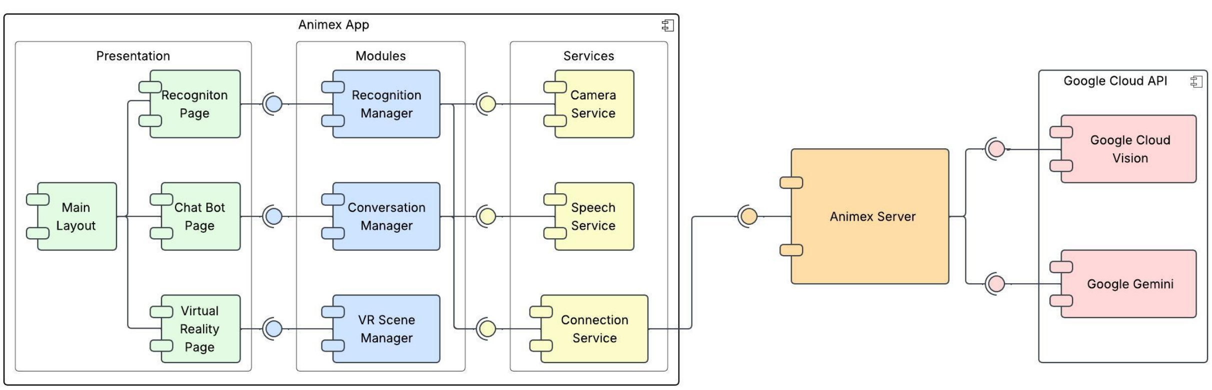
Given the aims of the MITA course, this report will focus on the architecture of the front-end application.

4.1 Front-end Application

4.1.1 Overview

The Animex application was developed utilizing the *Typescript* programming language, the *Vue*, *Ionic*, and *Three.js* graphical frameworks, and the *Capacitor* input framework.
The application itself can be divided into multiple layers:

- **Presentation Layer**: displays information to the user.
- **Module Layer**: handles the behavior needed for the system's tasks.
- **Service Layer**: interact with the user's device to provide the information and functions requested by the modules.



4.1.2 Service Layer

The service layer is composed of three services:

- **Connection Service**: connects to the back-end server, sends HTTP requests and forwards the responses to the appropriate modules.
- **Camera Service**: exposes the device's camera feed.
- **Speech Service**: connects to the user's microphone to convert their speech to text and reads out system messages through text-to-speech.

4.1.3 Module Layer

The module layer is formed by three modules, one for each of the system's tasks:

- **Recognition Manager**: periodically samples the camera feed through the *Camera Service* and then sends the image to the back-end using the *Connection Service*. Once the server's response is received, it saves the recognized animal as the current conversation subject or, if multiple animals were detected, it prompts the user to select one.
- **Conversation Manager**: keeps track of the messages sent between the user and the system's LLM, utilizes the *Speech Service* to elaborate user input and system output, and provides quiz functionality.
- **VR Scene Manager**: renders the VR environment with the selected animal's model and background environment and handles the VR-entities and their behaviors.

4.1.4 Presentation Layer

The presentation layer is split across four views:

- **Main Layout**: is overlaid on top of the other views, handles navigation between pages.
- **Recognition Page**: shows the device's camera feed, highlights identified animals in frame and allows the user to ask questions and respond to quizzes via speech.
- **Chat Bot Page**: shows the message history regarding the currently selected animal, allows the user to ask questions and request quizzes both via speech and via text.
- **Virtual Reality Page**: shows the VR scene, allows users to interact with the 3D model by giving it food and to ask questions via speech.

5 Response to Heuristic Evaluation

After receiving feedback from the heuristic evaluation, improvements were made to the system. Here follows a table with the issues risen and the solutions found:

Where	Interaction Breakdown	Violated Principle	Solution	Sev.
Scanner to Assistant transition	After scanning an animal, the system does not provide a clear transition between the Scanner task and the Assistant task. While the animal is correctly recognized, users are not guided towards the next available interaction, making the different tasks appear disconnected.	H1 - Visibility of system status; H2 - Match between system and the real world; multimodal grounding of task flow	The app's flow made this issue negligible: the user scans an animal and is then free to either go and chat with the animal, or go to the VR; this choice of action is purposely left to the user, making any kind or automatic redirect unnecessary.	3
Multiple-animal recognition	The current design does not clearly specify how the system behaves when multiple animals are detected within the same camera frame. Automatically selecting one animal may lead to misunderstandings and incorrect follow-up interactions.	H5 - Error prevention; H9 - Error recovery; ambiguity handling; referent grounding	This has been fixed by giving the user a way to choose the animal when an ambiguous scenario appears.	4
Assistant interaction	During the Assistant interaction, there is no persistent visual element indicating which animal is currently being discussed. Once the recognition task is completed, users may lose track of the active referent, especially during longer conversations. This weakens multimodal grounding and may make the Assistant appear disconnected from the previous recognition task.	H1 - Visibility of system status; multimodal grounding of referents	The currently recognized animal is now always visible to the user regardless of the section of the app.	3
Assistant speech output	The assistant's spoken responses cannot currently be interrupted while they are being generated or played. Users must wait until the response finishes before continuing the interaction.	H3 - User control and freedom; interruptible grounding; turn-taking in conversational interaction	The assistant can now be correctly interrupted.	3

Where	**Interaction Breakdown**	**Violated Principle**	**Solution**	**Sev.**
Quiz access inside Assistant	The Quiz entry point is located at the top of the Assistant page. After a long conversation, users may need to scroll through a large amount of chat history before being able to start the quiz.	H6 - Recognition rather than recall; H7 - Flexibility and efficiency of use	The quiz entry point is now in a fixed position, making it more accessible to use.	2
Technical / development information exposed to users	Recognition or backend-related information useful for developers can appear meaningless or confusing to children, creating a mismatch between a playful educational app and technical system feedback.	H4 - Consistency and standards; H2 - Match between system and real world	Development related information is now correctly transformed into data that the end user can read.	2
Mobile layout and system navigation overlap	On mobile devices, the chat input bar or bottom controls may be covered by Android/iOS navigation elements or by the keyboard. This can block the main action precisely when the user wants to ask a question or answer a quiz.	H4 - Consistency and standards; H7 - Efficiency of use; accessibility principles	The UI has been redone from scratch to effectively solve this issue and improve the user's overall experience.	3
Tutorial, onboarding and contextual help	The relationship between Scanner, Assistant and AR functionalities is not immediately clear. New users may not understand that animal recognition is expected before using the other functionalities. Furthermore, the available actions inside each task are not always self-evident.	H6 - Recognition rather than recall; H10 - Help and documentation	The new UI makes this kind of information clearer: when no animal is recognized, the other actions are locked, making it clear what the user has to do. after a recognition, said actions are unlocked.	2

Where	Interaction Breakdown	Violated Principle	Solution	Sev.
Overall application architecture	Although the application includes multiple modalities such as voice interaction, image recognition and AR, these modalities currently operate as separate tasks. The Assistant relies only on voice interaction, the Scanner relies only on image recognition, and the AR experience is isolated from the other functionalities. No task combines modalities simultaneously. As a consequence, the application is multimodal at the system level but not at the interaction level.	H2 - Match between system and real world; multimodal integration principles; grounding principles	More modalities have been added to the app: for example, now all the pages allow for the use of the voice chat, making the interaction truly multimodal.	4
Quiz generation	The quiz is generated dynamically through an LLM. It is unclear whether generated questions are always restricted to the information presented to the user about the selected animal. Additionally, LLM-generated quizzes may produce repetitive question structures over time, reducing educational variety.	H5 - Error prevention; educational consistency	The AI has been improved to give less repetitive questions and to be contextualized.	3